

Mayanka Medhe

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RESEARCH INTERESTS

Human-Robot Interaction, Vision-Language Models, Scene Understanding, Egocentric Vision

EDUCATION

University of Tokyo

Masters in Information Science & Technology

Japan

2020 - 2022

Thesis: Multimodal understanding of videos using eye gaze and natural language

Guide: [Prof. Yoichi Sato](#) in collaboration with **RIKEN CBS**

Indian Institute Of Technology, Bombay

BTech. with Honors in Computer Science and Engineering

India

2016 - 2020

Thesis: Multi-agent path planning using reinforcement learning

Guide: [Prof. Leena Vachhani](#)

Simon Fraser University

Semester Exchange majoring in Computing Science

Canada

Spring 2019

RESEARCH EXPERIENCE

Multimodal understanding of videos using eye gaze and natural language [\[THESIS\]](#)

University of Tokyo

Graduate Thesis / Guide: [Prof. Yoichi Sato](#)

Autumn 2020 - Spring 2022

Gaze-guided dense video captioning and multimodal dataset for egocentric attention understanding

- Proposed spatially and temporally dense video captioning task that generate time-aligned caption-gaze pairs
- Created a multimodal video dataset of gaze-aligned natural-language descriptions (2.5 hrs, 33 users, 15 languages)
- Integrated gaze priors into I3D features using attention maps & fixation-based encodings for dense captioning model
- Established enhanced action grounding via gaze-modulated features with active viewers gaze achieving 24.4% higher accuracy than passive viewers

Multi-Agent Path Planning and Collision Avoidance using Reinforcement Learning

IIT Bombay

Undergraduate Thesis / Guide: [Prof. Leena Vachhani](#)

Autumn 2019 - Spring 2020

Developed a path planning algorithm with collision detection & avoidance techniques

- Implemented a CNN-encoded DDQN policy for adaptive multi-agent path planning in dynamic settings
- Designed a variance-aware spherical safety zone for robust collision avoidance under velocity uncertainty
- Validated the framework in Gazebo with moving and stationary obstacles demonstrating reliable navigation

Playing ATARI games using Deep Learning Transformers [\[PAPER\]](#)[\[CODE\]](#)

IIT Bombay

Guide: [Prof. Shivaram Kalyanakrishnan](#)

Autumn 2019

Utilized transformers with deep neural networks for reinforcement learning tasks

- Proposed transformer based model for the task of training agents to play ATARI games using deep RL
- Utilized multi-head attention to encode temporal dependencies and extract richer state representations
- Evaluated RL performance under partial observability using DQN, DQN+LSTM, and transformer policies
- Found recurrent policies generalize better under hidden-state uncertainty than attention-only models

PROFESSIONAL EXPERIENCE

Honda R&D

Tokyo, Japan

Research Engineer, Human-Robot Interaction Team

2022 - Current

Embodied multimodal AI and human-robot interaction for autonomous micro-mobility

- Led smartwatch-robot interaction system enabling vision-language based control in real-world environments
- Built secure ROS-websocket communication and intent recognition for robust voice-driven operation
- Deployed system in public [proof-of-concept \(PoC\)](#) with 3000+ users handling varied real-world conditions
- Showcased our last-mile mobility Human-Robot Interaction system at the IFAC 2023 World Congress

Real-time perception & spatial intelligence for user identification and navigation

- Developed fine-grained pedestrian attribute recognition for user-aware autonomous mobility
- Optimized object detection performance on edge hardware, increasing robustness during real-world operation
- Building a Spatial RAG module enabling geospatial scene retrieval from vague natural language queries

Honda Research Institute

Research Engineer, Vision Team

Wako, Japan

Summer 2018

Multi-resolution 3D mapping using Dynamic Level of Detail for the bipedal robot ASIMO

- Constructed indoor 3D volumetric maps from SLAM point clouds, integrating ORB-SLAM2 for efficient mapping
- Optimized map reconstruction using feature-based feedback for efficient multi-resolution world modeling

Tier IV

Research Engineer, Sensing & Perception Team

Tokyo, Japan

Spring - Autumn 2021

Object detection evaluation metric design in self driving cars

- Designed and implemented object detection performance evaluation scheme in different weather conditions
- Implemented ROS-based module to detect false positive and false negative readings in object detection

Qibitech

Software Engineer, AI Team

Tokyo, Japan

Spring 2022

Highly autonomous remote control service (HATS) for efficient operation of robots

- Developed sub-modules for path planning of autonomous robots through teleoperation in a dynamic environment
- Created tests for localization and modification of cost map modules on various robots using Docker

PUBLICATIONS & PATENTS

- **Mayanka Medhe**, Naoki Hosomi. "User Localization in Camera Space using Combined User and Surrounding Object Information". JP Patent filed: Status Pending, 2025.
- **Mayanka Medhe**, Anirudh Reddy Kondapally, Kentaro Yamada, Hitomi Yanaka. "Improving Reasoning for Outdoor Navigation using Commonsense Knowledge Insertion". In: Society of Instrument and Control Engineers Annual Conference (SICE), 2023.
- Uddeshya Upadhyay*, Nikunj Shah*, Sucheta Ravikanti*, **Mayanka Medhe***. "Transformer Based Reinforcement Learning For Games". In: arXiv, 2019.

* work involved equal contribution from authors

COURSEWORK

Visual Computing	Computer Vision, Computer Graphics, Physics based animation
Machine Learning	Foundations of Intelligent and Learning Agents, NLP, Deep Learning
Miscellaneous	Advanced Topics in Mobile Robotics, Brain Science, Cognition & Emotion

TECHNICAL SKILLS

Programming	C++, C, Java, Python, Bash, SQL
Development	iOS, Android, HTML, CSS, Bootstrap, JavaScript, Django, PHP
Software & Platform	ROS2, Docker, OpenCV, OpenGL, Jenkins, Blender, Unity, Qt, Matlab, Git

ACHIEVEMENTS

- Received Honda R&D **Executive Award** for work on multimodal HRI for next-generation mobility (2025)
- Achieved **perfect GPA of 4.0/4.0** and selected as the graduate school representative at University of Tokyo (2022)
- Recipient of university recommended **Monbukagakusho (MEXT) scholarship** at University of Tokyo (2020-2022)
- Ranked **All India Rank 3** in National Science Talent Search Examination (2016)

COMPETITION

Design Engineer

Team Shunya- Pioneer for building zero energy house in India

IIT Bombay

2017 - 2018

- Only team from India to be selected for **Solar Decathlon China 2018** winning Best Participation Award
- Designed an Automated Hybrid Solar Convection Oven which uses Solar Hot Air Generator to cook food

EXTRACURRICULAR

- **Moderator:** Guided the winning team in the first Online Research Hackathon conducted by UTokyo (2021)
- **Freelance Instructor:** Conducted interactive coding and robotics workshops for kids aged 5-14 (2021-2022)
- **Volunteer:** Organized empowerment and English speaking programs at high schools across Japan (2021-2022)
- **Institute Student Mentor:** Assisted freshmen with academic and holistic endeavours (2019-2020)
- **Department Student Mentor:** Helped sophomores transition to department curriculum (2019-2020)